

TANUT BUMRUNGVONGSIRI

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SUMMARY

Artificial Intelligence engineer with 2+ years of hands-on experience designing, fine-tuning, and deploying AI solutions that automate complex workflows and improve productivity. Strong expertise in applying state-of-the-art models to real-world scenarios, with a focus on data characteristics, system constraints, and business requirements.

EDUCATION

King Mongkut's University of Technology Thonburi

July 2020 – May 2024

Bachelor's in Robotics and Automation Engineering, Institute of Field Robotics

Bangkok, Thailand

- GPAX: 3.74 / 4.00 (First-Class Honors)
- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Pattern Recognition, Natural Language Processing, Reinforcement Learning, Image Processing, Calculus, Statistics & Probability

EXPERIENCE

AI/ML Engineer

July 2024 – Present

Vulcorn AI · Full-time

Bangkok, Thailand

Digitalization for 2D Technical Engineering Drawings

- Designed and implemented an end-to-end pipeline for extracting textual and semantic information from 2D technical engineering drawings.
- Applied fine-tuned transformer-based OCR models to achieve high-accuracy text detection and recognition across complex drawing layouts.
- Leveraged fine-tuned Vision-Language Models (VLMs) to semantically understand and extract key metadata from 2D technical engineering drawings.

Digital Rebranding and Standardization for 2D Technical Drawings

- Implemented an automated pipeline to mask sensitive title block metadata in 2D engineering drawings and replace it with standardized, company-branded layouts.
- Utilized AI-based OCR and vision models to extract essential metadata while ensuring sensitive and proprietary information is protected.

AI/ML Engineer & Software Developer

June 2023 – November 2023

The Bangkok Unitrade Co., Ltd. · 6-month Internship

Bangkok, Thailand

- Designed and adapted object detection for surgical instruments moving along a conveyor belt for classification, employing YOLOv8.
- Conducted research and applied a multi-view classification model utilizing data from three camera perspectives to fine-grained classify surgical instruments, achieving 99.4% accuracy on 44 surgical instrument classes.
- Developed a software application with a PyQt5 graphical user interface (GUI) integrating AI-powered object detection and classification.
- Implemented logging functionality to record predictions in a MySQL database.

TECHNICAL SKILLS

Programming Languages: Python (Advanced); C, C++ (Intermediate)

Frameworks: PyTorch, TensorFlow, Keras, OpenCV, Transformers (Hugging Face), Ultralytics, scikit-learn

Backend & Deployment: FastAPI, Docker, Linux

Applied AI & ML: Model fine-tuning (including VLM-LoRA), OCR pipelines, computer vision systems, image detection and classification, end-to-end model deployment

PROJECTS

Vehicle Tracking using YOLOv8 + SORT

June 2024

- Leveraged YOLOv8 for vehicle detection and implemented tracking with SORT and Kalman filters for accurate estimation and tracking.

Deep Learning-based Navigation for Mobile Robots in Dynamic Environments

May 2024

- Adapted TD3 and SAC deep reinforcement learning algorithms for mobile robot navigation in dynamic environments with moving obstacles in Gazebo simulation.
- Grouped LiDAR clusters and tracked obstacles in each frame using the Hungarian Algorithm and Kalman Filter.

Solving Tic-Tac-Toe with Reinforcement Learning

April 2024

- Trained an agent to play Tic-Tac-Toe using Q-Learning, SARSA, and Double Q-Learning, and compared the performance of the three algorithms.

Image Classification of Surgical Instruments

March 2023

- Compared performance of fine-tuned ResNet-50 and VGG-16 models pre-trained on ImageNet, using data augmentation.
- Integrated the Attention Network mechanism with SEResNet50, enhancing the model's capability to focus on relevant features.
- Adapted fine-grained classification techniques (Bilinear Pooling, WSDAN) from research papers, achieving 98.5% accuracy on 4 surgical instrument classes.

Telemanipulation of Robot Hand using Human Gesture

November 2022

- Adapted MediaPipe and OpenCV to track 4 human gestures for controlling a universal robot arm and gripper.
- Implemented socket-based client-server communication to enable telemanipulation over the network.

Solving the 8-Puzzle and Mazes with AI

October 2022

- Developed Pygame software utilizing the A* algorithm to solve the 8-puzzle with the minimum number of moves, and adapted A* search to maze pathfinding problems.

PUBLICATION

Automated Bounding Dimension Retrieval: A Machine Learning-Driven Framework for Engineering Drawing Interpretation — ICCI 2025 (Co-author)

CERTIFICATION

Deep Learning for Image Classification — learn.thairobotics

June 2024

You Only Look Once (YOLO): Unified, Real-Time Object Detection — learn.thairobotics

June 2024

Image Processing — learn.thairobotics

November 2022